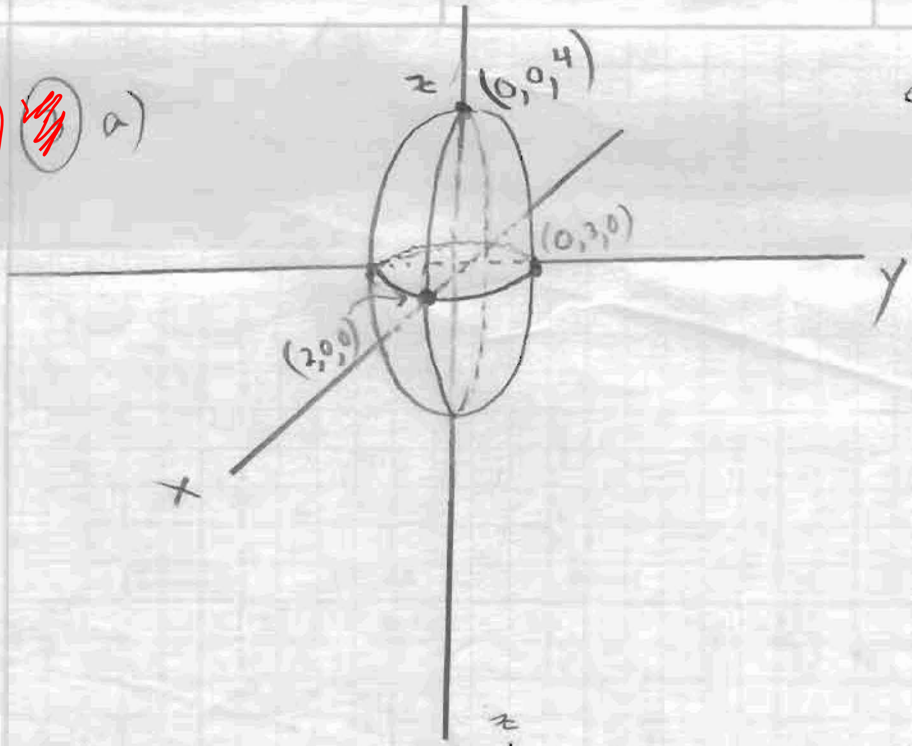
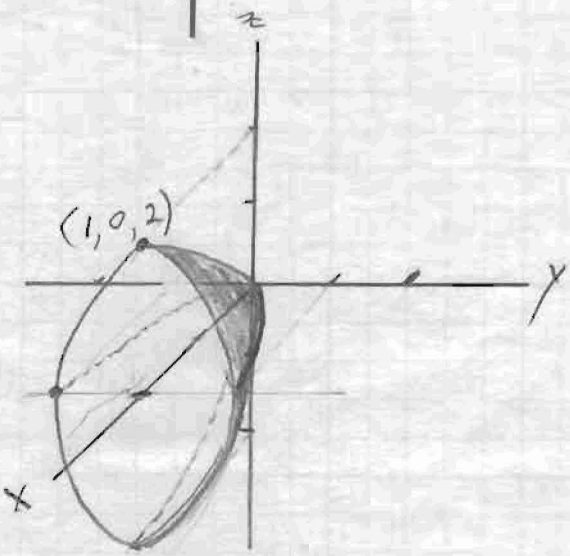
1

a)



ellipsoid

b)

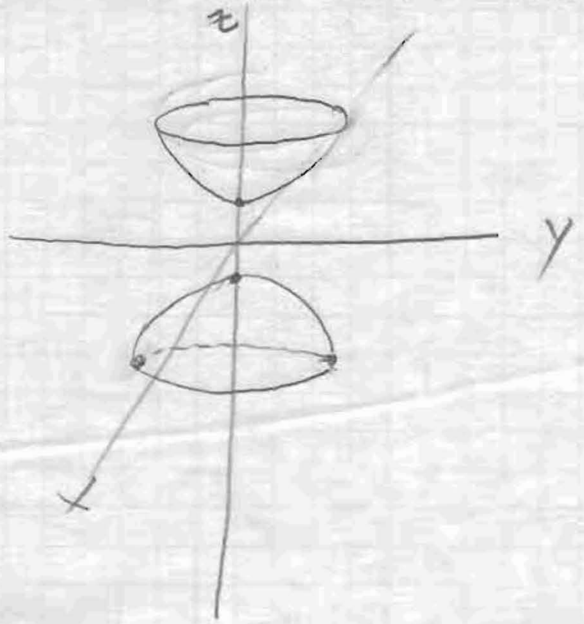


paraboloid

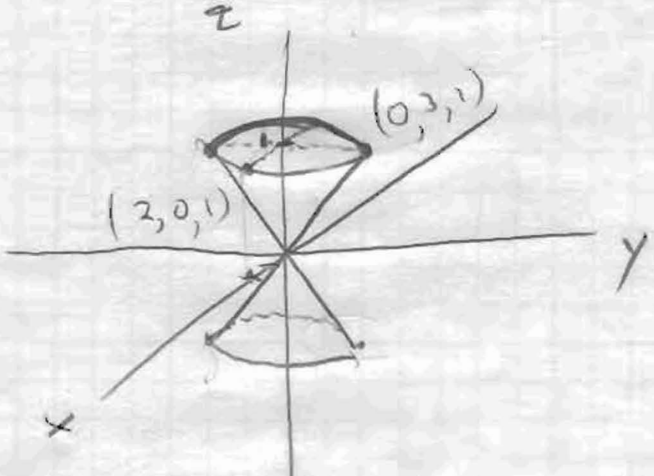
c) $z^2 = 1 + \frac{x^2}{4} + \frac{y^2}{9}$

$$z^2 - \frac{x^2}{4} - \frac{y^2}{9} = 1$$

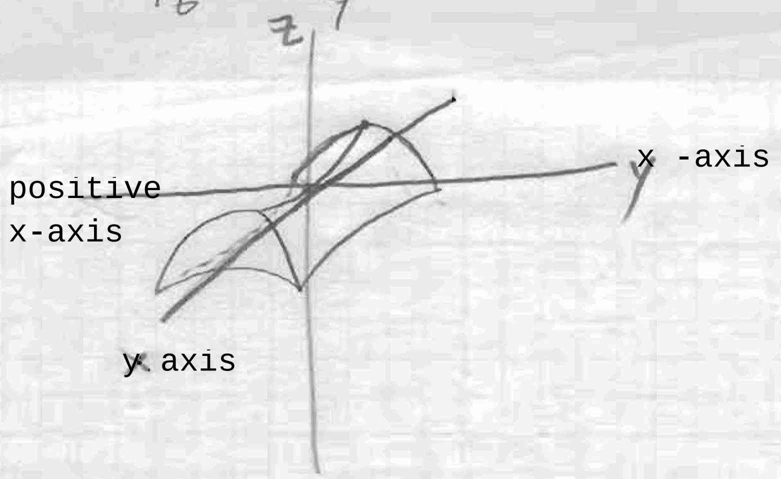
hyperboloid of two sheets



d) $z^2 = \frac{x^2}{4} + \frac{y^2}{9}$ cone

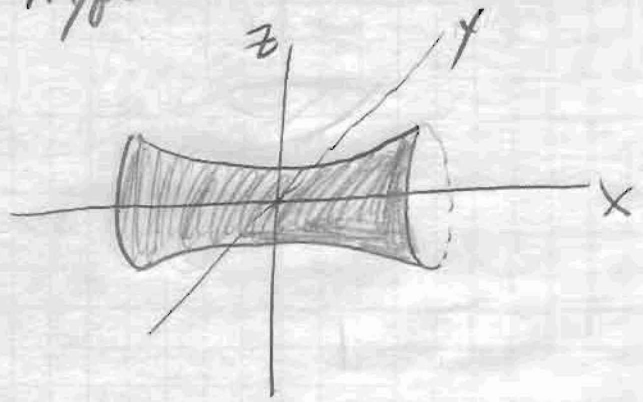


e) $z = \frac{y^2}{16} - \frac{x^2}{9}$ hyperbolic paraboloid



f) $y^2 + \frac{z^2}{9} - x^2 = 1$

hyperboloid of one sheet



④ a) $y^2 + 4y + 4x + z^2 - 10z + 29 = 0$

$y^2 + 4y + 4 + 4x + z^2 - 10z + 25 = 0$

$(y+2)^2 + 4x + (z-5)^2 = 0$

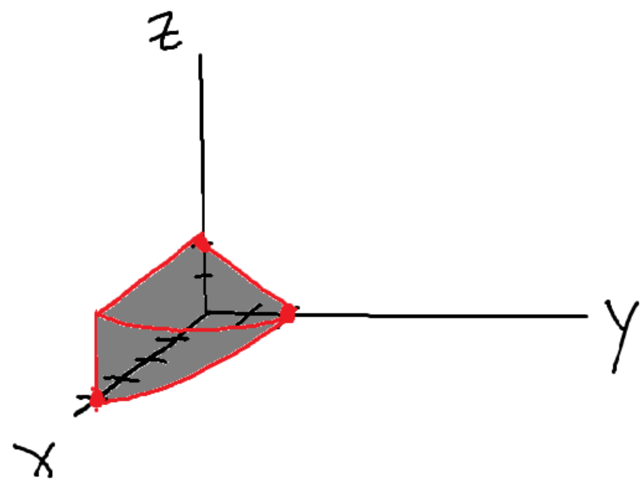
$4x = -\left[(y+2)^2 + (z-5)^2\right]$

$x = -\left[\frac{(y+2)^2}{4} + \frac{(z-5)^2}{4}\right]$ Elliptic Paraboloid

b) $(x-4)^2 + (y+2)^2 + (y-7)^2 = 25$

Sphere (ellipsoid) with center $(4,-2,7)$ and $r=5$

③



④